BIOGEOCHEMICAL CYCLES AND ECOLOGICAL PYRAMIDS

CYCLE ACCOUNTING FRAME → RESERVOIR CLASSIFICATION → CARBON MOVEMENT → NITROGEN TRANSFORMATION RAIL → NITROGEN ORGANISM MAP → PHOSPHORUS PATH

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CYCLE ACCOUNTING FRAME

CYCLE ACCOUNTING FRAME | SYSTEM BOUNDARY | ATMOSPHERE, WATER, SOIL, SEDIMENT AND LIVING BIOMASS INCLUDED | AMOUNT HELD IN ONE RESERVOIR AT A STATED TIME | TRANSFER BETWEEN RESERVOIRS PER STATED TIME | BIOLOGICAL, CHEMICAL OR GEOLOGICAL TRANSFORMATION | INPUTS MINUS OUTPUTS; NEVER ASSUMED FROM ONE ISOLATED STOCK

SYSTEM BOUNDARY → atmosphere, water, soil, sediment and living biomass included → POOL → amount held in one reservoir at a stated time → FLUX → transfer between reservoirs per stated time → PROCESS → biological, chemical or geological transformation → BALANCE → inputs minus outputs; never assumed from one isolated stock

SYSTEM INPUT / STARTING CONDITION
- SYSTEM BOUNDARY → atmosphere, water, soil, sediment and living biomass included
- POOL → amount held in one reservoir at a stated time

ECOLOGICAL PROCESS / TRANSFER
- FLUX → transfer between reservoirs per stated time
- PROCESS → biological, chemical or geological transformation

OUTCOME / FEEDBACK / LIMIT
- BALANCE → inputs minus outputs; never assumed from one isolated stock
- SYSTEM BOUNDARY → atmosphere, water, soil, sediment and living biomass included

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BALANCE → inputs minus outputs; never assumed from one isolated stock.

01

STAGE 01 RESERVOIR CLASSIFICATION

RESERVOIR CLASSIFICATION | MAJOR ATMOSPHERE OR OCEAN RESERVOIR | MAJOR CRUST, SOIL OR SEDIMENT RESERVOIR | CARBON AND NITROGEN | PROMINENT ATMOSPHERIC EXCHANGE | ROCK-WEATHERING AND SEDIMENT PATHWAY | CLASS DOES NOT SET ONE UNIVERSAL CYCLING SPEED

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
GASEOUS → major atmosphere or ocean reservoir	CARBON AND NITROGEN → prominent atmospheric exchange	CAUTION → class does not set one universal cycling speed
SEDIMENTARY → major crust, soil or sediment reservoir	PHOSPHORUS → rock-weathering and sediment pathway	GASEOUS → major atmosphere or ocean reservoir

ECOLOGICAL UNIT / PARAMETER
- GASEOUS → major atmosphere or ocean reservoir
- SEDIMENTARY → major crust, soil or sediment reservoir

DECISIVE DISTINCTION
- CARBON AND NITROGEN → prominent atmospheric exchange
- PHOSPHORUS → rock-weathering and sediment pathway

SCALE / STATUS / EXCEPTION
- CAUTION → class does not set one universal cycling speed
- GASEOUS → major atmosphere or ocean reservoir

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → class does not set one universal cycling speed.

02

STAGE 02 CARBON MOVEMENT

CARBON MOVEMENT | ATMOSPHERIC OR DISSOLVED CARBON | PHOTOSYNTHETIC FIXATION | PRODUCER BIOMASS | CONSUMERS AND DETRITAL PATHWAYS | CARBON RETURNED DURING METABOLISM | DECOMPOSITION AND COMBUSTION | ADDITIONAL RETURN PATHWAYS

ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation → PRODUCER BIOMASS → consumers and detrital pathways → RESPIRATION → carbon returned during metabolism → DECOMPOSITION AND COMBUSTION → additional return pathways → RULE → stored carbon and annual transfer are different measures

SYSTEM INPUT / STARTING CONDITION
- ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation
- PRODUCER BIOMASS → consumers and detrital pathways

ECOLOGICAL PROCESS / TRANSFER
- RESPIRATION → carbon returned during metabolism
- DECOMPOSITION AND COMBUSTION → additional return pathways

OUTCOME / FEEDBACK / LIMIT
- RULE → stored carbon and annual transfer are different measures
- ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → stored carbon and annual transfer are different measures.

03

STAGE 03 NITROGEN TRANSFORMATION RAIL

NITROGEN TRANSFORMATION RAIL | ATMOSPHERIC NITROGEN | AMMONIA OR USABLE NITROGEN FORMS | ORGANIC NITROGEN | PLANT UPTAKE OR DENITRIFICATION | RETURN TOWARD ATMOSPHERIC NITROGEN

ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms → ORGANIC NITROGEN → ammonification → ammonia → AMMONIA → nitrification → nitrite → nitrate → NITRATE → plant uptake or denitrification → DENITRIFICATION → return toward atmospheric nitrogen

SYSTEM INPUT / STARTING CONDITION
- ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ECOLOGICAL PROCESS / TRANSFER
- AMMONIA → nitrification → nitrite → nitrate

OUTCOME / FEEDBACK / LIMIT
- DENITRIFICATION → return toward atmospheric nitrogen

- ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation
- PRODUCER BIOMASS → consumers and detrital pathways
- RESPIRATION → carbon returned during metabolism
- DECOMPOSITION AND COMBUSTION → additional return pathways
- RULE → stored carbon and annual transfer are different measures
- ATMOSPHERIC OR DISSOLVED CARBON → photosynthetic fixation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → stored carbon and annual transfer are different measures.

03

STAGE 03 NITROGEN TRANSFORMATION RAIL

NITROGEN TRANSFORMATION RAIL | ATMOSPHERIC NITROGEN | AMMONIA OR USABLE NITROGEN FORMS | ORGANIC NITROGEN | PLANT UPTAKE OR DENITRIFICATION | RETURN TOWARD ATMOSPHERIC NITROGEN

ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms → ORGANIC NITROGEN → ammonification → ammonia → AMMONIA → nitrification → nitrite → nitrate → NITRATE → plant uptake or denitrification → DENITRIFICATION → return toward atmospheric nitrogen

SYSTEM INPUT / STARTING CONDITION
- ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms
- ORGANIC NITROGEN → ammonification → ammonia

ECOLOGICAL PROCESS / TRANSFER
- AMMONIA → nitrification → nitrite → nitrate
- NITRATE → plant uptake or denitrification

OUTCOME / FEEDBACK / LIMIT
- DENITRIFICATION → return toward atmospheric nitrogen
- ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ATMOSPHERIC NITROGEN → fixation → ammonia or usable nitrogen forms.

04

STAGE 04 NITROGEN ORGANISM MAP

NITROGEN ORGANISM MAP | SYMBIOTIC FIXATION ASSOCIATED WITH LEGUME ROOTS | FREE-LIVING FIXATION EXAMPLE IN THE OWNER | AMMONIA TO NITRITE STEP | NITRITE TO NITRATE STEP | DENITRIFICATION ROUTE

RHIZOBIUM → symbiotic fixation associated with legume roots

AZOTOBACTER → free-living fixation example in the owner → NITROSOMONAS → ammonia to nitrite step → NITROBACTER → nitrite to nitrate step

SYSTEM / LEVEL / DEFINITION
- RHIZOBIUM → symbiotic fixation associated with legume roots
- AZOTOBACTER → free-living fixation example in the owner

STRUCTURE-FUNCTION MECHANISM
- NITROSOMONAS → ammonia to nitrite step
- NITROBACTER → nitrite to nitrate step

SCALE / EVIDENCE LIMIT
- PSEUDOMONAS-TYPE → denitrification route
- RHIZOBIUM → symbiotic fixation associated with legume roots

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AZOTOBACTER → free-living fixation example in the owner.

05

STAGE 05 PHOSPHORUS PATH

PHOSPHORUS PATH | PHOSPHATE ROCK | SOIL OR WATER PHOSPHATE | PLANT UPTAKE | CONSUMER TRANSFER | DETRITAL RETURN | AQUATIC LOADING | MINING AND FERTILISER

PHOSPHATE ROCK → weathering → soil or water phosphate → PLANT UPTAKE → consumer transfer → detrital return → RUNOFF → aquatic loading → sedimentation → MINING AND FERTILISER → accelerate extraction and dispersal → BOUNDARY → no significant atmospheric phase in the owner

SYSTEM / LEVEL / DEFINITION
- PHOSPHATE ROCK → weathering → soil or water phosphate
- PLANT UPTAKE → consumer transfer → detrital return

STRUCTURE-FUNCTION MECHANISM
- RUNOFF → aquatic loading → sedimentation
- MINING AND FERTILISER → accelerate extraction and dispersal

SCALE / EVIDENCE LIMIT
- BOUNDARY → no significant atmospheric phase in the owner
- PHOSPHATE ROCK → weathering → soil or water phosphate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → no significant atmospheric phase in the owner.

06

STAGE 06 CYCLE-SPECIFIC HUMAN PRESSURE

CYCLE-SPECIFIC HUMAN PRESSURE | FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE | SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN | MINING, FERTILISER APPLICATION AND RUNOFF | CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER | SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CARBON → fossil-fuel combustion and land-cover change	PHOSPHORUS → mining, fertiliser application and runoff	POLICY → source control must match the cycle and reservoir
NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen	OUTCOMES → climate forcing, nutrient pollution and resource loss differ	CARBON → fossil-fuel combustion and land-cover change

ECOLOGICAL UNIT / PARAMETER
- CARBON → fossil-fuel combustion and land-cover change
- NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

DECISIVE DISTINCTION
- PHOSPHORUS → mining, fertiliser application and runoff
- OUTCOMES → climate forcing, nutrient pollution and resource loss differ

SCALE / STATUS / EXCEPTION
- POLICY → source control must match the cycle and reservoir
- CARBON → fossil-fuel combustion and land-cover change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → no significant atmospheric phase in the owner.

06

STAGE 06

CYCLE-SPECIFIC HUMAN PRESSURE

CYCLE-SPECIFIC HUMAN PRESSURE · FOSSIL-FUEL COMBUSTION AND LAND-COVER CHANGE · SYNTHETIC FERTILISER AND COMBUSTION-RELATED REACTIVE NITROGEN · MINING, FERTILISER APPLICATION AND RUNOFF · CLIMATE FORCING, NUTRIENT POLLUTION AND RESOURCE LOSS DIFFER · SOURCE CONTROL MUST MATCH THE CYCLE AND RESERVOIR

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CARBON → fossil-fuel combustion and land-cover change	PHOSPHORUS → mining, fertiliser application and runoff	POLICY → source control must match the cycle and reservoir
NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen	OUTCOMES → climate forcing, nutrient pollution and resource loss differ	CARBON → fossil-fuel combustion and land-cover change

ECOLOGICAL UNIT / PARAMETER

- CARBON → fossil-fuel combustion and land-cover change
- NITROGEN → synthetic fertiliser and combustion-related reactive nitrogen

DECISIVE DISTINCTION

- PHOSPHORUS → mining, fertiliser application and runoff
- OUTCOMES → climate forcing, nutrient pollution and resource loss differ

SCALE / STATUS / EXCEPTION

- POLICY → source control must match the cycle and reservoir
- CARBON → fossil-fuel combustion and land-cover change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → source control must match the cycle and reservoir.

07

STAGE 07

EUTROPHICATION MECHANISM

EUTROPHICATION MECHANISM · NUTRIENT ENRICHMENT · INCREASED ALGAL OR PLANT PRODUCTION · ORGANIC MATERIAL ACCUMULATES · DECOMPOSER ACTIVITY RISES · OXYGEN DEMAND RISES · DISSOLVED OXYGEN MAY DECLINE · ECOLOGICAL STRESS

NUTRIENT ENRICHMENT → increased algal or plant production → ORGANIC MATERIAL ACCUMULATES → decomposer activity rises → OXYGEN DEMAND RISES → dissolved oxygen may decline → ECOLOGICAL STRESS → composition changes and mortality may occur → CLASSIFY → natural ageing versus human-accelerated cultural eutrophication

SYSTEM INPUT / STARTING CONDITION

- NUTRIENT ENRICHMENT → increased algal or plant production
- ORGANIC MATERIAL ACCUMULATES → decomposer activity rises

ECOLOGICAL PROCESS / TRANSFER

- OXYGEN DEMAND RISES → dissolved oxygen may decline
- ECOLOGICAL STRESS → composition changes and mortality may occur

OUTCOME / FEEDBACK / LIMIT

- CLASSIFY → natural ageing versus human-accelerated cultural eutrophication
- NUTRIENT ENRICHMENT → increased algal or plant production

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLASSIFY → natural ageing versus human-accelerated cultural eutrophication.

08

STAGE 08

THREE PYRAMID PARAMETERS

THREE PYRAMID PARAMETERS · COUNT OF ORGANISMS AT EACH TROPHIC LEVEL · STANDING LIVING MATERIAL AT A STATED TIME · TRANSFER THROUGH TROPHIC LEVELS OVER TIME · DEPENDS ON PARAMETER AND ECOSYSTEM · ONLY THE ENERGY PYRAMID IS ALWAYS UPRIGHT

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NUMBERS → count of organisms at each trophic level	ENERGY → transfer through trophic levels over time	RULE → only the energy pyramid is always upright
BIOMASS → standing living material at a stated time	SHAPE → depends on parameter and ecosystem	NUMBERS → count of organisms at each trophic level

ECOLOGICAL UNIT / PARAMETER

- NUMBERS → count of organisms at each trophic level
- BIOMASS → standing living material at a stated time

DECISIVE DISTINCTION

- ENERGY → transfer through trophic levels over time
- SHAPE → depends on parameter and ecosystem

SCALE / STATUS / EXCEPTION

- RULE → only the energy pyramid is always upright
- NUMBERS → count of organisms at each trophic level

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → only the energy pyramid is always upright.

09

STAGE 09

AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION · SMALL PHYTOPLANKTON STANDING CROP · RAPID REPLACEMENT · HIGH TURNOVER · CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD · CONSUMER STANDING BIOMASS · MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT · ENERGY TRANSFER

SMALL PHYTOPLANKTON STANDING CROP → rapid replacement → HIGH TURNOVER → continued production over the measurement period → CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant → ENERGY TRANSFER → still declines upward → VERDICT → biomass inversion is not energy inversion

SYSTEM INPUT / STARTING CONDITION

- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement
- HIGH TURNOVER → continued production over the measurement period

ECOLOGICAL PROCESS / TRANSFER

- CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant
- ENERGY TRANSFER → still declines upward

OUTCOME / FEEDBACK / LIMIT

- VERDICT → biomass inversion is not energy inversion
- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

09

STAGE 09
AQUATIC BIOMASS INVERSION

AQUATIC BIOMASS INVERSION · SMALL PHYTOPLANKTON STANDING CROP · RAPID REPLACEMENT · HIGH TURNOVER · CONTINUED PRODUCTION OVER THE MEASUREMENT PERIOD · CONSUMER STANDING BIOMASS · MAY EXCEED PRODUCER BIOMASS AT ONE INSTANT · ENERGY TRANSFER

SMALL PHYTOPLANKTON STANDING CROP → rapid replacement → HIGH TURNOVER → continued production over the measurement period → CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant → ENERGY TRANSFER → still declines upward → VERDICT → biomass inversion is not energy inversion

SYSTEM INPUT / STARTING CONDITION
- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement
- HIGH TURNOVER → continued production over the measurement period

ECOLOGICAL PROCESS / TRANSFER
- CONSUMER STANDING BIOMASS → may exceed producer biomass at one instant
- ENERGY TRANSFER → still declines upward

OUTCOME / FEEDBACK / LIMIT
- VERDICT → biomass inversion is not energy inversion
- SMALL PHYTOPLANKTON STANDING CROP → rapid replacement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → biomass inversion is not energy inversion.

10

STAGE 10
HEURISTIC AND MODEL FIREWALL

HEURISTIC AND MODEL FIREWALL · ENERGY LOSS · QUALITATIVE THERMODYNAMIC DIRECTION IS ROBUST · TEN-PER-CENT RULE · OWNER LABELS IT AN AVERAGE HEURISTIC · WEAKENS A RIGID ONE-LEVEL ASSIGNMENT · CHANGES STANDING STOCKS AND FEEDING RELATIONS · USEFUL SUMMARY, NOT A COMPLETE FOOD WEB

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
ENERGY LOSS → qualitative thermodynamic direction is robust	OMNIVORY → weakens a rigid one-level assignment	PYRAMID → useful summary, not a complete food web
TEN-PER-CENT RULE → owner labels it an average heuristic	SEASONALITY → changes standing stocks and feeding relations	ENERGY LOSS → qualitative thermodynamic direction is robust

ECOLOGICAL UNIT / PARAMETER
- ENERGY LOSS → qualitative thermodynamic direction is robust
- TEN-PER-CENT RULE → owner labels it an average heuristic

DECISIVE DISTINCTION
- OMNIVORY → weakens a rigid one-level assignment
- SEASONALITY → changes standing stocks and feeding relations

SCALE / STATUS / EXCEPTION
- PYRAMID → useful summary, not a complete food web
- ENERGY LOSS → qualitative thermodynamic direction is robust

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEN-PER-CENT RULE → owner labels it an average heuristic.

11

STAGE 11
PYQ AND EVIDENCE BOUNDARY

PYQ AND EVIDENCE BOUNDARY · ROUTED CONCEPTS · AGRICULTURAL NITROGEN, ROCK-WEATHERED PHOSPHORUS, FIXATION LINKS · BASIC PRACTICE · RESERVOIR, PROCESS AND ORGANISM DISTINCTIONS · OFFICIAL PAPER · DEMAND WORDING ONLY · NO KEY

ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links → BASIC PRACTICE → reservoir, process and organism distinctions → OFFICIAL PAPER → demand wording only → NO KEY → no option or answer inferred → LIVE CHECK → no rate, pool, efficiency or status imported from a stub

ECOLOGICAL UNIT / PARAMETER
- ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links
- BASIC PRACTICE → reservoir, process and organism distinctions

DECISIVE DISTINCTION
- OFFICIAL PAPER → demand wording only
- NO KEY → no option or answer inferred

SCALE / STATUS / EXCEPTION
- LIVE CHECK → no rate, pool, efficiency or status imported from a stub
- ROUTED CONCEPTS → agricultural nitrogen, rock-weathered phosphorus, fixation links

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIVE CHECK → no rate, pool, efficiency or status imported from a stub.

E

EXTRA
SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · CPCB MONITORS WATER-QUALITY PARAMETERS (BOD, DISSOLVED OXYGEN, NUTRIENT LOAD) · MINISTRY OF AGRICULTURE AND FARMERS WELFARE · DEPARTMENT OF FERTILIZERS · IMPLEMENTATION GAP · FERTILISER SUBSIDY DESIGN IN INDIA HAS HISTORICALLY FAVOURED · CARBON-CYCLE SCIENCE (IPCC AR6, 2023) IS THE SCIENTIFIC SPINE · NATIONAL-SCALE NUTRIENT-CYCLE DISRUPTION (E.G., TOTAL AGRICULTURAL NITROGEN RUNOFF) IS

OPTIONAL ECOLOGICAL PROCESS DEPTH
- CPCB monitors water-quality parameters (BOD, dissolved oxygen, nutrient load) that
- Ministry of Agriculture and Farmers Welfare / Department of Fertilizers: regulate
- CAUTION: Implementation gap: fertiliser subsidy design in India has historically favoured
- CAUTION: Carbon-cycle science (IPCC AR6, 2023) is the scientific spine of the entire UNFCCC/Paris

SCALE, PARAMETER OR STATUS LIMIT
- CAUTION: National-scale nutrient-cycle disruption (e.g., total agricultural nitrogen runoff) is
- CAUTION: "Ecological efficiency of 10%" is an average heuristic; actual transfer efficiency
- Gaseous cycles (carbon, nitrogen, oxygen, water) self-regulate faster than sedimentary
- Biological nitrogen fixation converts atmospheric N_2 to ammonia; industrial fixation via

COMPARATIVE RESTORATION NUANCE
- The pyramid of energy is always upright due to progressive energy loss at each trophic
- Eutrophication depletes dissolved oxygen as decomposer microbial activity spikes to
- NOT: Nutrient pollution and carbon emissions are governed by the same policy lever. - They
- NOT: An inverted biomass pyramid means an unhealthy ecosystem. - It reflects fast nutrient

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels